Building ML/AI Applications

Introduction and administration

Carlos Cotrini and the teaching team

CAS ETH in AI and Software Development *D-INFK, ETH Zürich*

Autumn semester 2026

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Section 1

About us

About Carlos

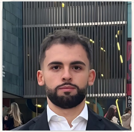
Dr. Carlos Cotrini

- BSc Mathematics, 2009
- MSc Information Security, ETH, 2014
- PhD Information Security, ETH, 2019
- PostDoc in AI and ML, ETH, 2019 to 2021
- Senior scientist, ETH, 2021 to now

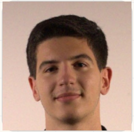
Courses at ETH and UZH

- Advanced machine learning, 2019 to now
- Statistical learning theory, 2020 to 2022
- AI in medicine, 2020 to 2022
- Data analysis in physics, 2020 to 2025
- Stochastics and machine learning, 2024 to now
- Building AI/ML applications, 2024 to now
- AI and IT in industry, 2025 to now
- Software engineering fundamentals, 2025 to now
- AI project, 2026 to now
- Applied introduction to machine learning, from 2027

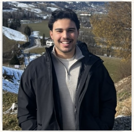
The teaching team



Ivan Angjelovski

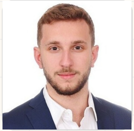


Amine Benjelloun Touimi



Ghali Berbich

Francesco Caperna



Konstantinos Chasiotis

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Suei-Wen Chen

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Luca Chimienti

Monica De Alvaro Mena

Mattia de Martino

The teaching team

Photo wanted

Vassili De Palma

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Hassan El Bouz

Alper Karadeniz

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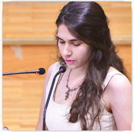
Joel Kaufmann



Matina Lampaki

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Angelo Marano

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Samy Messouak

Elisej Orsini-Rosenberg

Frederick Puls

The teaching team

Photo wanted

Kangsheng Qi

Photo wanted

Lamberto Ragnolini

Winston Rohmann

Daniele Russica

Sanziana Stelea

Christina Tsakanika

Photo wanted

Celal Turkmen

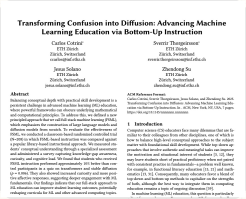
Photo wanted

Jinlei Zhou

We study how we teach this

Transforming Confusion into Diffusion SIGCSE TS 2026
CER Best Paper (Global) award

- **Full stack ML**: you build the model from the ground up before you reach for a library.
- Trial with $N = 208$: that beat the library first approach by about 10 percent on conceptual understanding ($p = 0.006$), and left participants more curious.



Transforming Confusion into Diffusion: Advancing Machine Learning Education via Bottom-Up Instruction

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Abstract

Balancing conceptual depth with practical skill development is a persistent challenge in advanced machine learning (ML) education, where powerful frameworks can obscure underlying mathematical and computational principles. To address this, we defined a new principled approach that we call full-stack machine learning (FSML), which emphasizes the construction of large language models and diffusion models from scratch. To evaluate the effectiveness of FSML, we conducted a classroom-based randomized controlled trial (N=208) in which FSML-based instruction was compared against a popular library-based instructional approach. We measured students' conceptual understanding through a specialized assessment and administered a survey capturing knowledge-gap awareness, curiosity, and cognitive load. We found that students who received FSML instruction performed approximately 10% better than control participants in a quiz on transformers and stable diffusion ($p = 0.006$). They also showed increased curiosity and more positive affective responses, suggesting deeper engagement with ML fundamentals. Our findings indicate that our full-stack approach to ML education can improve student learning outcomes, potentially reshaping curricula for ML and other advanced computing topics.

ACM Reference Format:
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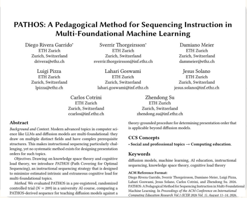
1 Introduction

Computer science (CS) educators face many dilemmas that are familiar to their colleagues from other disciplines, one of which is how to balance high-level conceptual approaches to the subject matter with foundational skill development. While top-down approaches that involve authentic and meaningful tasks can improve the motivation and situational interest of students [3, 12], they may leave students short of practical proficiency when not paired with consistent practice in fundamentals—a problem well known, for example, in functional literacy education [15, 21] and mathematics [13, 31]. Consequently, many educators favor a blend of top-down and bottom-up methods to capitalize on the strengths of both, although the best way to integrate them in computing education remains a topic of ongoing discussion [39].

In machine learning (ML) education, this question is particularly

PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning ICER 2026

- *"…multi-foundational: they draw on multiple distinct fields and have complex prerequisite structures."*
- The order we teach things in is itself designed. Trial with $N = 209$: the designed order scored higher ($p = 0.011$, $d = 0.32$) at lower cognitive load.



PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning

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Abstract

Background and Context. Modern advanced topics in computer science like LLMs and diffusion models are multi-foundational: they draw on multiple distinct fields and have complex prerequisite structures. This makes instructional sequencing particularly challenging, yet no systematic method exists for designing presentation orders for such topics.

Objectives. Drawing on knowledge space theory and cognitive load theory, we introduce *PATHOS* (Path Covering for Optimal Sequencing), an instructional sequencing strategy that is designed to minimize estimated intrinsic and extraneous cognitive load for multi-foundational topics.

Method. We evaluated PATHOS in a pre-registered, randomized controlled trial ($N = 209$) in a university AI course, comparing a PATHOS-derived sequence for teaching diffusion models against a

theory-grounded procedure for determining presentation order that is applicable beyond diffusion models.

CCS Concepts

• **Social and professional topics → Computing education.**

Keywords

diffusion models, machine learning, AI education, instructional sequencing, knowledge space theory, cognitive load theory

ACM Reference Format:
Diego Rivera Garrido, Sverrir Thorgeirsson, Damiano Meier, Luigi Pizza, Lahari Goswami, Jesus Solano, Carlos Cotrini, and Zhendong Su. 2026. PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning. In *Proceedings of the ACM Conference on International Computing Education Research Vol.1 (ICER 2026 Vol. 1), August 11–14, 2026.*

Experimental-study team Dr. Sverrir Thorgeirsson · Stefan Cadar · Sofi Mousia · Umut Demirbas

Johannes Schulte-Vels · Nikolaj Murasov · Jonathan Maillefaud · Oscar Jauffret

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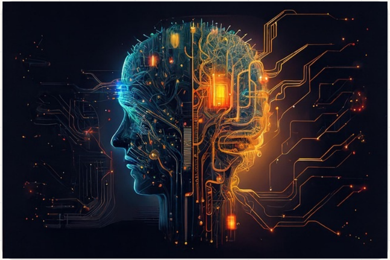
Section 2

Artificial intelligence, what is it?

Four paradigms in science

	Ontological as it should be	**Epistemic** as it is
Thinking with brains	Probability 	Statistics 
Computing with computers	Computer science 	**Machine learning** 

Artificial intelligence, a working definition



The design and implementation of systems that perform tasks normally requiring human intelligence.

Main approaches

- **Symbolic reasoning**: rules and logic
- **Machine learning**: estimate dependencies from data
- **Deep learning**: machine learning with deep neural networks

Artificial intelligence, a timeline

Foundations


Turing test, 1950


Dartmouth conference, 1956


Rosenblatt's perceptron, 1958

Expert systems, AI winter

LISP, 1958

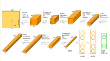
DENDRAL and MYCIN, 1970s

Statistical machine learning


Deep Blue, 1997


IBM Watson, 2011

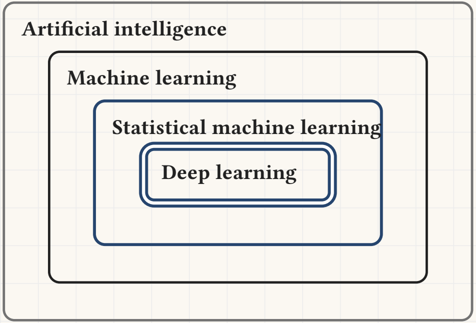
Deep learning


AlexNet, 2012


AlphaGo, 2016

ChatGPT, 2022

From artificial intelligence to deep learning



- **Artificial intelligence** designs programs that act like intelligent systems, by whichever means.
- **Machine learning** does it by estimating dependencies from observed data.
- **Statistical machine learning** estimates those dependencies with Vapnik's empirical risk minimisation.
- **Deep learning** does the same with deep neural networks.

Building ML applications, and hence AI applications.

You learn to use deep learning to design applications that create value from data.

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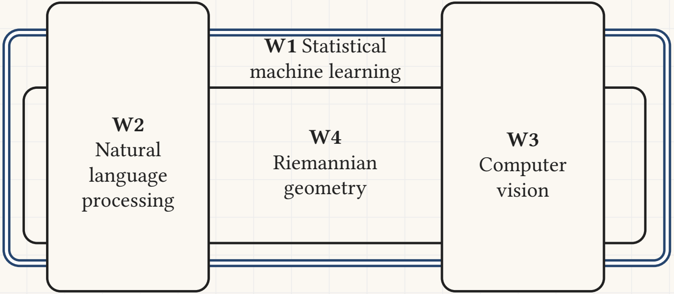
Section 3

Scope and learning goals

What you learn in this course

This course goes deeper into the machine learning you met earlier in the programme. We explore four areas of modern AI.

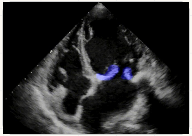


For each area you build an application: a piece of software that uses AI in an innovative way to produce business value.

Some success stories

Oliver Krone, 2025, head of grid and hydro engineering at BKW.

- Entered the Kaggle competition of Advanced machine learning, where students segment the mitral valve of human hearts from echocardiogram videos. Placed 12th out of 100.

The FDD cohort, 2026, a whole class of working professionals.

- Handed the first graded project of *Advanced Machine Learning*, an ETH Master's course, with no scaffolding.
- Their 18 teams beat its 136 Master's teams on mean, median and consistency. Top score: an AML grade of **5.95**.

Bernhard Obrist, 2025, senior project manager at Aargauische Kantonalbank.

- Built an agent that learns on its own to drive a two dimensional road. Carlos's own agent crashes after 15 seconds.
- Demo: `https://polybox.ethz.ch/index.php/s/raoa3SBoG7cPnT2`


A whole class of professionals, against an ETH Master's course

	AML ($n = 136$)	FDD ($n = 18$)
Mean	0.7115	**0.7426**
Median	0.7164	**0.7420**
Std	0.0485	**0.0309**
Min	0.4865	**0.6774**
Max	**0.8062**	0.7922

- Participants of our sister course were handed the first graded project of *Advanced Machine Learning*, an ETH Master's course, with no project structure to lean on.
- Their 18 teams beat its 136 Master's teams on mean, median and consistency. Only **one** AML team beat their best score.
- They did it in **12.6** submissions per team against the Master's students' **25.6**. Their top score converts to an AML grade of **5.95**.

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Section 4

Administration

Please read these slides carefully.

The course website

`https://eth-bmai-hs26.github.io/BMAI-PAGE/`

- The schedule for all four weekends, Friday and Saturday.
- The coding exercises for every session.
- The project descriptions and their grading schemes.

Scan it to open the site on your phone

Passing the course

To pass the course you must do all of the following.

1. Complete the first three of the four applications and submit your code on Moodle: Moodle course link
2. Defend each of those three in a 15 minute presentation to one of the TAs.
3. Pass three of the four Moodle quizzes. You may retake a quiz as often as you need, but you must reach at least 80 percent of the points.

Everything above must be finished before the grand deadline

Sunday 25 October 2026, 23:59

The applications

- Each application is tied to one of the four areas above. We release one after each weekend, in order. **Four are offered, three are required.**
- We recommend that you finish each application within one month.
- Applications are solved **in pairs** wherever possible.
- Form your pair, choose the TA you will present to, and record that choice: TA choice and defence sign up
- For each application, arrange a 15 minute Zoom meeting with that TA and defend it before the grand deadline.

Weekend	**Application**
W1, Vapnik's statistical learning theory	LLM routing
W2, Natural language processing	Tax agent
W3, Computer vision	Fashion magazine editor
W4, Riemannian geometry	Deep fake generation and detection

The four applications

One application per weekend, each one a piece of software that uses AI to produce business value. **The first three are required.**

W1 LLM router

Statistical machine learning

Send each question to the cheapest model that can still answer it well.

W2 Tax agent

Natural language processing

Read tax documents and answer questions about them in plain language.

W3 Fashion magazine editor

Computer vision

Produce the images from text: describe the shot, get the picture.

W4 Deep fake generation and detection

Riemannian geometry

Build the generator, then build the detector that catches it.

Google Colab

- We recommend Google Colab for this course: `https://colab.research.google.com/`
- You get a Google Colab Pro licence, valid until the grand deadline.
- To claim it you first set up a Google workspace. Instructions: unlimited.ethz.ch, Google Workspace, first steps

Office hours

- The teaching team holds office hours over Zoom between the weekends.
- Drop in and ask the TAs about the lecture or about your applications.
- Days and times: Office hours days and times
- Zoom link: Office hours Zoom link
- Sign up for a slot: Office hours sign up link

Questions?

eth-bmai-hs26.github.io/BMAI-PAGE